

① x^2 在左: 加切减对, 开方移位.

$$\int \frac{dx}{x^2+a^2} = \frac{1}{a} \arctan \frac{x}{a} + C. \text{ "加切" } \rightarrow \text{加法变为正切}$$

$$\int \frac{dx}{x^2-a^2} = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C \quad / \quad \int \frac{dx}{a^2-x^2} = \frac{1}{2a} \ln \left| \frac{x+a}{x-a} \right| + C \quad \text{("减对" } \rightarrow \text{减法变为对数)}$$

(互为倒数)

$$\int \frac{dx}{\sqrt{x^2+a^2}} = \ln(\sqrt{x^2+a^2} + x) + C \quad \text{"开方移位" } \rightarrow \text{开平方则移位上方再加 } x \text{ 放入对数}$$

$$\int \sqrt{x^2+a^2} = \frac{x}{2} \sqrt{x^2+a^2} + \frac{a^2}{2} \ln(\sqrt{x^2+a^2} + x) + C \quad \leftarrow \text{ (这个是自加的 } \sim \text{ 与另一个记忆方法类似)}$$

② x^2 在右: 下弦 翻身另加 $\frac{x}{2}$ 倍

$$\int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin \frac{x}{a} + C. \text{ (下方变正弦)}$$

$$\int \sqrt{a^2-x^2} = \frac{x}{2} \sqrt{a^2-x^2} + \frac{a^2}{2} \arcsin \frac{x}{a} + C$$

③ 三角: ① 出现分母想对数, 化简首 (要添负) ($\csc x, \cot x$ 首字母为 c)

$$\int \tan x dx = \int \frac{\sin x}{\cos x} dx = -\ln |\cos x| + C$$

$$\int \cot x dx = \int \frac{\cos x}{\sin x} dx = \ln |\sin x| + C$$

$$\int \sec x dx = \ln |\sec x + \tan x| + C. \quad \left. \begin{array}{l} \text{记忆: } (\tan x)' = \sec^2 x \\ (\cot x)' = -\csc^2 x \end{array} \right\} \text{ 一家人放在一起}$$

$$\int \csc x dx = -\ln |\csc x + \cot x| + C. \quad \left. \begin{array}{l} \text{记忆: } (\tan x)' = \sec^2 x \\ (\cot x)' = -\csc^2 x \end{array} \right\}$$

$$(\ln |\csc x - \cot x| + C = \ln |\tan \frac{x}{2}| + C)$$

② 二次降次留分母, 弦1互换判正负.

1. $\int \sec^2 x dx = \int \frac{1}{\cos^2 x} dx = \frac{\sin x}{\cos x} + C = \tan x + C$

2. $\int \csc^2 x dx = \int \frac{1}{\sin^2 x} dx = \frac{\cos x}{\sin x} + C = -\cot x + C$

3. $\int \sec x \tan x dx = \int \frac{\sin x}{\cos^2 x} dx = \frac{1}{\cos x} + C = \sec x + C$

4. $\int \csc x \cot x dx = \int \frac{\cos x}{\sin^2 x} dx = -\frac{1}{\sin x} + C = -\csc x + C$

“1”与一个分母不同的弦互换
二次项, 降次后留在分母上
“1”换正弦为正, 换余弦为负.
(记忆: $(\cos x)'$ 是负的)

变限积分求导问题

$$F(x) = \int_{\psi(x)}^{\varphi(x)} f(t) dt, \quad F'(x) = f(\varphi(x)) \varphi'(x) - f(x, \psi(x)) \psi'(x)$$

$$F(x) = \int_{\psi(x)}^{\varphi(x)} f(x, t) dt = \int_{\psi(x)}^{\varphi(x)} f_x(x, t) dt + f(x, \varphi(x)) \varphi'(x) - f(x, \psi(x)) \psi'(x)$$

当 $a > |b|$ 时, $\int_0^z \frac{1}{a + b \cos x} dx = \frac{z}{\sqrt{a^2 - b^2}}$